MATH 1075 - Section 10.1, 10.2 Worksheet

11/3/2026

1 Square Roots

Evaluate the roots.

1. $\sqrt{25}$

2. $-\sqrt{25}$

3. $\sqrt{-25}$

4. $\sqrt{\frac{100}{121}}$

5. $\sqrt{0.81}$

6. $\sqrt{0.16}$

2 n'th Roots

Evaluate the roots.

1. $\sqrt[3]{64}$

2. $\sqrt[3]{-125}$

3. $\sqrt[3]{\frac{1}{8}}$

4. $\sqrt[4]{-1}$

5. $\sqrt[3]{-\frac{8}{27}}$

3 Roots of Variable Expressions

Simplify the radical expressions.

1. $\sqrt{(x+1)^2}$

2. $\sqrt{x^2y^4}$

3. $\sqrt[3]{(u+v)^3}$

4. $\sqrt[4]{\frac{a^4}{b^8}}, b \neq 0$

4 Rational Exponents

Write the expression in radical form and simplify.

1. $49^{1/2}$

2. $49^{-1/2}$

3. $-16^{1/4}$

4. $(-16)^{1/4}$

5 Converting Between Rational Exponents and Radical Notation

Convert each expression to radical notation.

1. $t^{3/5}$

2. $6y^{3/4}$

3. $x^{2/3}y^{1/3}$

4. $6r^{-2/5}$

Write each expression by using rational exponents.

1. $\sqrt[3]{x}$

2. $10\sqrt{b}$

3. $\sqrt[3]{y^2}$

4. $\sqrt[4]{a^2b^3}$

6 Properties of Rational Exponents

Simplify the expressions by using the properties of rational exponents.

1. $x^{1/4}x^{-5/4}$

2. $\frac{x^{5/3}}{x^{2/3}}$

3. $(y^{1/5})^{10}$

4. $(a^{1/3}a^{1/4})^{12}$